

Decoding the Noise: Deep Learning for RF Modulation

Redefining Telecommunications with High-Precision CNN Architecture.

In the rapidly evolving landscape of cognitive radio and telecommunications, distinguishing signal types in noisy environments is critical. This project leverages the RadioML dataset to construct a robust Convolutional Neural Network (CNN) capable of Automatic Modulation Recognition (AMR).

Deep Learning Architecture:

Designed a multi-layer CNN model specifically tailored to process raw IQ (In-Phase and Quadrature) samples, eliminating the need for manual feature extraction.

Performance Under Noise:

Optimized the model to classify complex modulation schemes—including BPSK, QPSK, and 8PSK—achieving superior accuracy even in low Signal-to-Noise Ratio (SNR) scenarios where traditional methods fail.

Optimization & Tuning:

Implemented rigorous data preprocessing pipelines and hyperparameter tuning to prevent overfitting, demonstrating the practical scalability of AI in next-gen signal processing.

